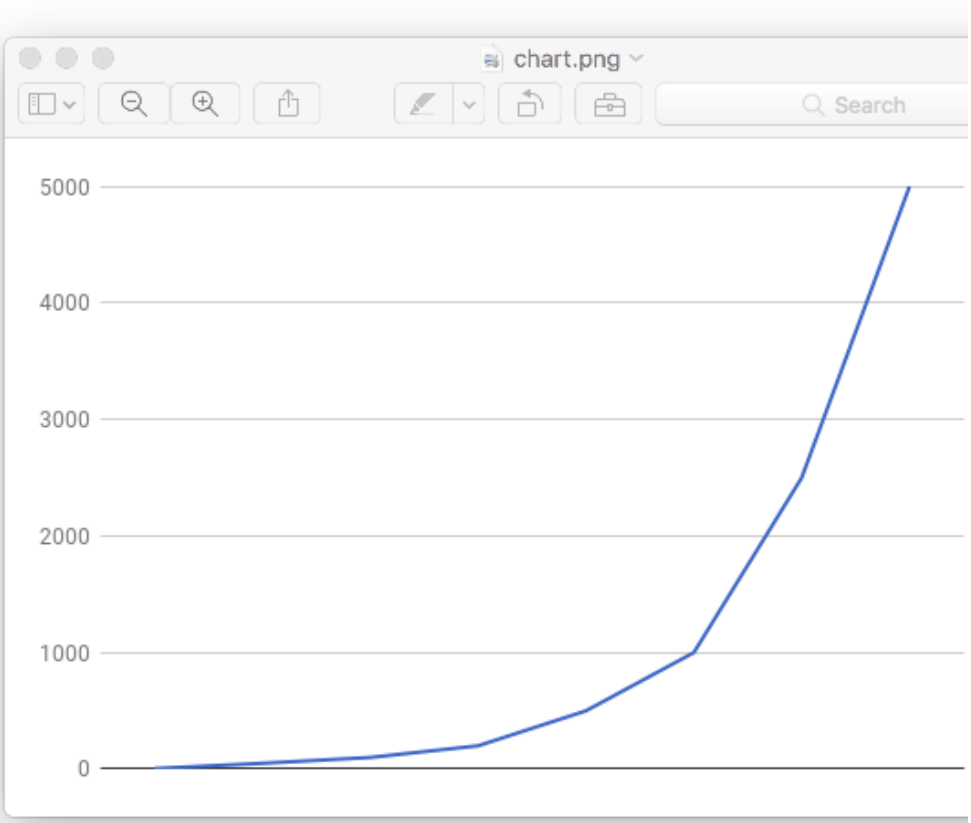
Torry Brelsford

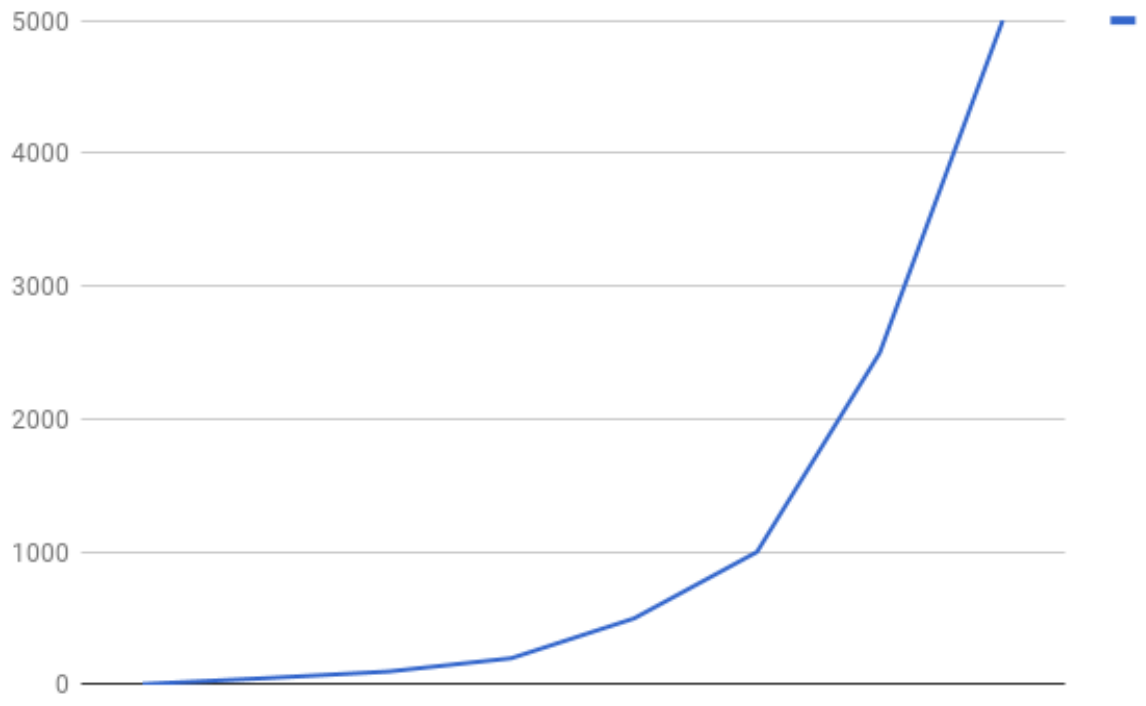
Include an explanation of the asymptotic analysis of the function, and state your reason based on the algorithm that you have implemented.

- I have a few issues with the time complexity of my function. I am calling compute_in_degrees after every node I delete with a zero indegree and recomputing each nodes indegrees. The function compute_in_degrees has a n^2 complexity as does my own compute_tps for a n^4 which is fine for smaller values but as n gets to 2500+ the time it takes to complete is really terrible. Although it works correctly, it does not run with the desired speed. Some things I could have done to improve the time complexity would be to decrement the in-degree dictionary when the node I am deleting has an edge to those nodes. This would take away the need to run compute_in_degrees after every deleted node from the graph.

Time	Nodes

0.00093	5
0.0097	50
0.03589	100
0.17738	200
2.11182	500
21.2845	1000
388.59651	2500
3883.20198	5000




5000
4000
3000
2000
1000
0